



SIF Analysis and Retrieval

Author: Carmen Oliver Huidobro (VU)

Supervisors: Gerbrand Koren (UU) and Sandjai Bhulai (VU)

Department of Mathematics
Vrije Universiteit van Amsterdam
Amsterdam, The Netherlands

May 13, 2025

Preview of Master Thesis

Climate change is not a future event, it is already reshaping our planet [Pat+22]. Monitoring key environmental parameters, such as the Gross Primary Production (GPP) of the terrestrial biosphere, is essential for understanding the global carbon cycle and its impact on climate dynamics [Pia+13]. GPP is the rate at which solar energy is captured in sugar molecules during photosynthesis by living biomass and represents the largest gross flux of CO_2 between the atmosphere and Earth surface. The total GPP by terrestrial plants is around 122 billion tons per year; in comparison, burning fossil fuels emits about 7 billion tons annually [Bee+10]. Changes in climate conditions (like temperature and precipitation) affect how much CO_2 plants can absorb during photosynthesis. In turn, the amount of CO_2 in the atmosphere can influence the climate [Cha13], for example, by enhancing the greenhouse effect. This mutual influence is known as “carbon cycle–climate coupling”. Because these processes are so closely linked, uncertainties in one part affect the other. This coupling makes it challenging to accurately forecast future carbon uptake and its effects on the environment.

Solar-Induced Fluorescence (SIF) has emerged as a promising proxy for GPP [Fra+11]. SIF is a faint electromagnetic signal re-emitted by chlorophyll during photosynthesis. From the solar energy absorbed by vegetation, approximately 1–2% is released as fluorescence, around 19% as heat, and the remaining 80% is used for photosynthesis [Bak08] (see Fig.1).

SIF exhibits a smooth spectral signature, with a prominent far-red peak at 737 nm and a secondary red peak at 684 nm, each associated with distinct photosystems [Ped+08] (see Fig.2). SIF intensity depends on the balance between absorbed light and how efficiently it is used for photosynthesis—regulated by factors like CO_2 concentration, canopy temperature, water, and nutrient availability. Under high-light conditions, studies have shown a strong correlation between SIF and photosynthetic activity, linking it to both absorbed photosynthetically active radiation (APAR) and photosynthetic efficiency [Mia+18]. Therefore, despite the intricate nature of the underlying processes, satellite-derived far-red SIF is directly compared with GPP.

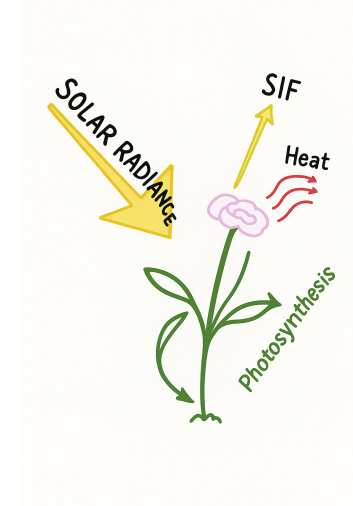


Figure 1: Graphic describing plant solar irradiance absorption and energy release after photosynthesis.

The Amazon forest is the largest tropical rain forest in the world, storing 40% of the global biomass of tropical forests [Mal+06]. Thus this region plays a crucial role in global climate regulation. The Amazon basin has typical tropical climate characteristics: warm temperatures (averaging 27.9°C in the dry season and 25.8°C in the rainy season) and high humidity (88% in the wet season, 77% in the dry season). This humidity increases atmospheric H_2O levels, making SIF retrieval particularly sensitive to choices in wavelength windows and atmospheric modeling [Van+20].

Despite logistical difficulties, in situ measurements such as those from the Amazon Tall Tower Observatory (ATTO) provide valuable localized insights. ATTO uses eddy covariance (EC) and partitioning methods to measure photosynthesis. However, EC measurements are spatially limited, typically representing an area of 1 km^2 and there almost none of them (see Fig 3). In contrast, the Amazon spans 6.7 million km^2 —twice the size of India—underscoring the need for scalable, satellite-based methods on retrieval of environmental parameters that serve as a good proxy for photosynthesis, such as SIF.

SIF is a faint signal emitted by plants, and detecting it from space requires very sensitive instruments. Satellite sensors such as ENVISAT [Gua+07], GOSAT [Gua+12], GOME-2 (v1 [San+16], v2 [Van+20], v3 [Ane+24]), SCIAMACHY [Joi+16], OCO-2 [Sun+18], and TROPOMI [Gua+21b; Köh+18] have all

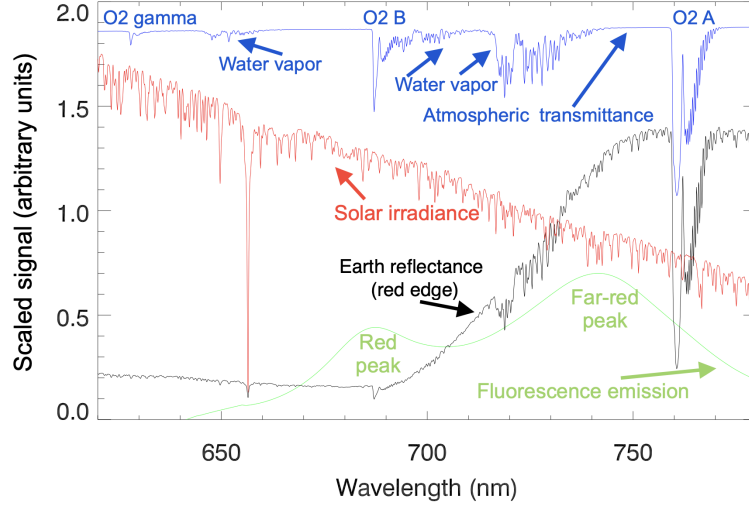


Figure 2: Simulated terrestrial spectra: atmospheric transmittance, reflectance, SIF emissions, and solar irradiance as a function of wavelength (from [Joi+16]).

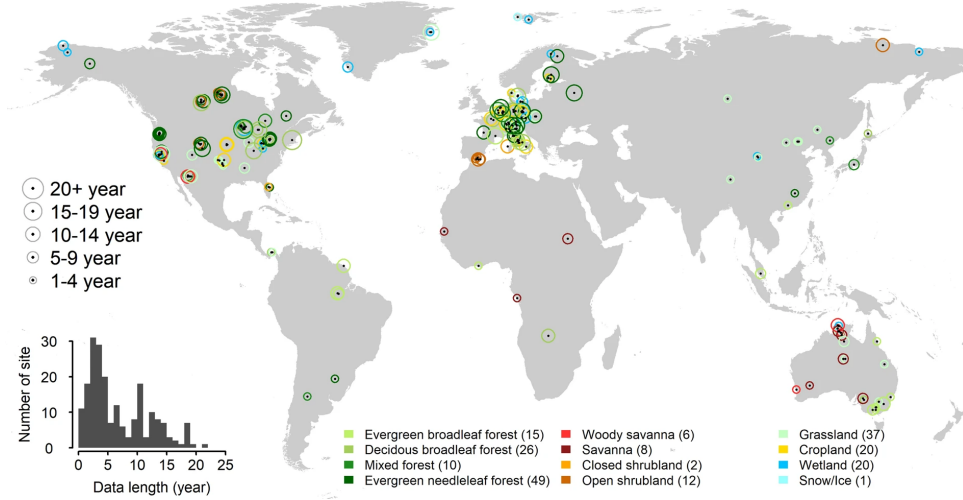


Figure 3: Map of 206 tower sites from the 212 sites in the February 2020 release of the FLUXNET2015 dataset. The size of the circle indicates the length of the data record. The color of the circles represents the ecosystem type based on the International Geosphere-Biosphere Programme (IGBP) definition. When overlapping, locations are offset slightly to improve readability. Numbers in parentheses indicate the number of sites in each IGBP group. The inset shows the distribution of data record lengths.[Pas+20].

been used to retrieve SIF. These sensors require high signal-to-noise ratios and fine spectral resolutions to capture the faint SIF signal. However, coarse resolution and low signal to noise ratio still limit the usability of data in small or fragmented ecosystems.

The release of TROPOMI aboard Sentinel-5P in 2018 marked a significant improvement, with a resolution as high as $7 \text{ km} \times 3.5 \text{ km}$, with daily morning overpasses. This high-quality data allows researchers to explore spatial and temporal trends in vegetation productivity [Gua+21a], [Gua+21b]. This is a great upgrade from past sensors such as GOME-2A that has a resolution of $40 \text{ km} \times 40 \text{ km}$, more than 5 times lower than TROPOMI. This sensor also had many complications such as sensitiveness to calibration issues, instrument degradation [Ane+24] and low resolution data.

SIF data has been used to detect correlations with extreme weather events such as droughts and floods. Studies suggest the Amazon could face biome shifts, biodiversity loss, and carbon depletion due to climate change (e.g., [NB09], [Sak+16]). There is still limited understanding of how elevated CO_2 , droughts, and warming affect ecosystem water-use efficiency and photosynthetic capacity [Ram+10].

Therefore measuring the impact of extreme weather events on the Amazon rainforest is of great importance to understand its vulnerability.

SIF serves as a valuable proxy for assessing photosynthetic activity and detecting plant stress, particularly under extreme climatic events. By analyzing spatiotemporal anomalies in SIF, researchers can gain insights into the impacts of phenomena such as droughts, floods, and fires on ecosystem productivity. To identify these anomalies, statistical methods like z-scores and median-based techniques are employed. These methods aim to filter out noise and emphasize unusual variations in SIF data. The relevance of SIF in monitoring plant stress is underscored by studies such as Koren et al. (2018) [Kor+18], which demonstrated that SIF was strongly suppressed over areas with anomalously high temperatures and decreased soil moisture during the 2015/2016 El Niño event. This suppression correlated with reduced GPP, highlighting SIF’s sensitivity to environmental stressors. By providing a direct observation of photosynthetic activity, SIF enables the assessment of how extreme weather events influence the carbon cycle, thereby informing conservation and mitigation strategies.

Satellite derived SIF data can originate from different sensors, each employing distinct retrieval algorithms. SIF is measured by analyzing the absorption features in the solar spectrum, particularly the Fraunhofer lines—dark, well-defined spectral bands across a broad wavelength range. When chlorophyll in vegetation absorbs sunlight, a fraction of this energy is re-emitted as fluorescence, slightly “filling in” these dark lines. By examining the red to near-infrared region (680–760 nm), where Fraunhofer lines are abundant, researchers can detect and quantify this filling effect, which corresponds to the faint fluorescence signal emitted by plants. This approach, pioneered by [Joi+13] and [KGJ15], enables remote assessment of photosynthetic activity with high spectral precision. However, the SIF signal is inherently weak, making its retrieval highly sensitive to initial assumptions.

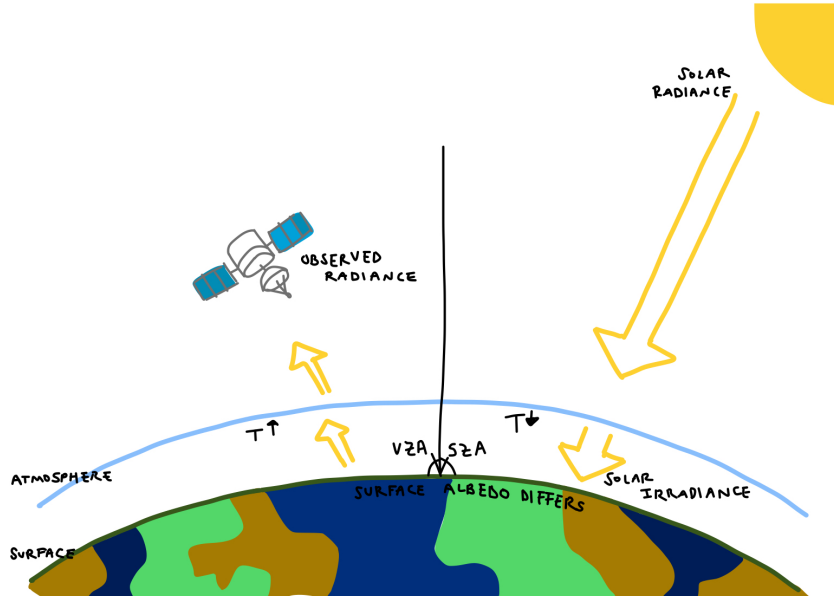


Figure 4: Graphic describing solar irradiance on earth and satellite’s observation: Solar angle, viewing angle, atmospheric transmittance and surface albedo all play a role in the observed radiance. See equation ?? for further understanding.

For instance, the GOME-2 sensor employs three variants of the SIFTER retrieval algorithm, differing in key technical aspects such as the wavelength window, atmospheric optical depth modeling, and surface albedo fitting order [Ane+24], [Van+20], [San+16]. Even minor adjustments in these parameters can alter the retrieved SIF values. The SIFTER algorithm models atmospheric transmittance using Principal Component (PC) analysis derived from reference spectra over fluorescence-free regions (e.g., the Sahara) (Fig. 5). However, the selection of these reference scenes, the statistical treatment of atmospheric transmittance, and the choice of fitting window can strongly influence the final SIF estimates [Ane+24].

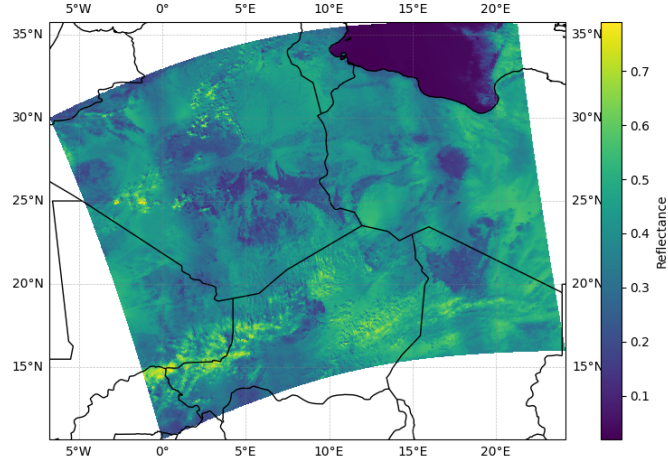


Figure 5: Reflectance observed by Copernicus Sentinel-5P TROPOMI mission over North and West Africa. Note that the reflectance shown here is measured by the high-resolution TROPOMI instrument whereas the SIFTER retrieval is so far based on the coarser GOME-2 instrument.

In this context, several key questions arise regarding SIF. Two research questions guide this study:

1. How can ML techniques help detect anomalies in SIF data over key carbon exchange regions like the Amazon? How can these anomalies inform us about the ecosystem's response to extreme weather events?
2. To what extent do small methodological differences in the SIFTER retrieval algorithm affect the resulting data? How robust are current assumptions used in the estimation of SIF from satellite observations?